

Individual #04 Add a new line-item UI Form. Add Reports.

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Deliverables

1. Screenshots

1. Receipt list view and form view. Show that Receipt is available from the menu, opens to list view with at least 5 records.

Receipts

+ Create New

Q Search...

1-5 of 5 receipt10 / page

RECEIPT NO	DATE	CUSTOMER NAME	PAYMENT METHOD	TOTAL RECEIVED	ACTIONS
RCT26-00005	April 21, 2026	Krungthep Trading Co., Ltd.	Cash	฿5,892.49	EditDelete
RCT26-00004	April 21, 2026	Thonburi Trading Co., Ltd.	Check	฿20,665.06	EditDelete
RCT26-00003	April 21, 2026	Siam Trading Co., Ltd.	Check	฿9,548.43	EditDelete
RCT26-00001	April 21, 2026	Placeholder Customer B	Bank Transfer	฿4,217.95	EditDelete
RCT26-00002	April 20, 2026	Placeholder Customer A	Cash	฿5,549.83	EditDelete

ภาพหน้าจอแสดงรายชื่อของ Receipt list อย่างน้อย 5 รายการ

Create Receipt

← Back

Receipt Information

Receipt No

NEW (Generated)

Date *

21/04/2026

Payment Method *

Cash

Billing Information

Customer *

Select a customer...Search

Payment Notes

Optional notes...

Applied Invoices

+ Add Unpaid Invoices

INVOICE NO	AMOUNT DUE	ALREADY RCVD	REMAINING	AMT RECEIVED HERE	STILL REMAINING	ACTION
Select a customer and add unpaid invoices to receive payment.						

Total Received: ฿0.00

Cancel

Save Receipt

ภาพหน้าจอแสดงหน้าของ Receipt form สำหรับการเพิ่ม/แก้ไขข้อมูลใบเสร็จ

2. Receipt form view. Show 2-3 receipt form views already saved with at least 2 invoices per receipt and with enough data to show that the line-item logic especially that related to “amount remaining” works correctly.

Receipt RCT26-00001

[← Back](#)[Edit](#)[Print PDF](#)

InvoiceDoc v2

Customer
C101 - Placeholder Customer B
2 Main St, -

OFFICIAL RECEIPT
Date: April 20, 2026
Receipt No: RCT26-00001
Payment Method: Bank Transfer

INVOICE NO	AMOUNT DUE	ALREADY RCVD	REMAINING	AMT RECEIVED HERE	STILL REMAINING
INV-012	฿813.07	฿0.00	฿813.07	฿813.07	฿0.00
INV-037	฿3,404.88	฿0.00	฿3,404.88	฿3,404.88	฿0.00

Thank you for your payment.

Total Received: ฿4,217.95

Receipt RCT26-00003

[← Back](#)[Edit](#)[Print PDF](#)

InvoiceDoc v2

Customer
C102 - Siam Trading Co., Ltd.
660 Rama IX Rd, Nonthaburi

OFFICIAL RECEIPT
Date: April 20, 2026
Receipt No: RCT26-00003
Payment Method: Check

INVOICE NO	AMOUNT DUE	ALREADY RCVD	REMAINING	AMT RECEIVED HERE	STILL REMAINING
INV-002	฿2,169.43	฿0.00	฿2,169.43	฿2,169.43	฿0.00
INV-013	฿1,407.32	฿0.00	฿1,407.32	฿1,407.32	฿0.00
INV-027	฿390.02	฿0.00	฿390.02	฿390.02	฿0.00
INV-035	฿1,000.32	฿0.00	฿1,000.32	฿1,000.32	฿0.00
INV-040	฿4,581.34	฿0.00	฿4,581.34	฿4,581.34	฿0.00

Thank you for your payment.

Total Received: ฿9,548.43

Receipt RCT26-00005

[← Back](#)[Edit](#)[Print PDF](#)

InvoiceDoc v2

Customer
C108 - Krungthep Trading Co., Ltd.
332 Bang Na-Trat Rd, Nonthaburi

OFFICIAL RECEIPT
Date: April 20, 2026
Receipt No: RCT26-00005
Payment Method: Cash

INVOICE NO	AMOUNT DUE	ALREADY RCVD	REMAINING	AMT RECEIVED HERE	STILL REMAINING
INV-032	฿2,096.93	฿0.00	฿2,096.93	฿2,096.93	฿0.00
INV-038	฿2,573.89	฿0.00	฿2,573.89	฿2,573.89	฿0.00
INV-050	฿1,221.67	฿0.00	฿1,221.67	฿1,221.67	฿0.00

Thank you for your payment.

Total Received: ฿5,892.49

ภาพหน้าจอแสดงหน้าจอ Receipt form ที่ถูกบันทึกไว้แล้ว 2-3 รายการ โดยในแต่ละใบเสร็จมีรายการใบแจ้งหนี้ที่รับชำระแล้วอย่างน้อย 2 รายการ

3. Receipt line-item Invoice LoV correctly filters to unpaid invoices for the selected customer only. Include a screenshot that proves this.

Edit Receipt RCT26-00001

← Back

Receipt Information

Receipt No

RCT26-00001

Date *

20/04/2026

Payment Method *

Bank Transfer

Billing Information

Customer *

C101 - Placeholder Customer B

Search

Payment Notes

Optional notes...

Applied Invoices

+ Add Unpaid Invoices

INVOICE NO	AMOUNT DUE	ALREADY RCVD	REMAINING	AMT RECEIVED HERE	STILL REMAINING	ACTION
INV-012	฿813.07	฿0.00	฿813.07	813.07	฿0.00	
INV-037	฿3,404.88	฿0.00	฿3,404.88	3404.88	฿0.00	

Total Received:฿4,217.95

Cancel Save Receipt

เมื่อกดปุ่ม Add Unpaid Invoices

Select Unpaid Invoices

×

Q Search Invoice No...

	INVOICE NO	AMT DUE	ALREADY RECEIVED	REMAINING
☐	INV-012	฿813.07	฿0.00	฿813.07
☐	INV-037	฿3,404.88	฿0.00	฿3,404.88
☐	INV-044	฿2,516.91	฿0.00	฿2,516.91
☐	INV-048	฿497.55	฿0.00	฿497.55

0 selected

Cancel Apply Selected

ภาพหน้าจอแสดงหน้าจอ Receipt form ของ LoV (List of Values) สำหรับใบแจ้งหนี้ จะต้องกรองเฉพาะใบแจ้งหนี้ที่ยังไม่ได้ชำระเงิน (Add Unpaid Invoices) สำหรับลูกค้ารายที่เลือกเท่านั้น (อย่างในภาพจะเลือก Customer B ซึ่งจะขึ้นใบแจ้งหนี้เฉพาะของ Customer B เท่านั้น)

4. Printed receipt screenshot for 1 receipt (choose a nice demo receipt with 2–3 line items).

22/4/69 00:01

InvoiceDoc v2

InvoiceDoc v2

Customer

C108 - Krungthep Trading Co., Ltd.
332 Bang Na-Trat Rd, Nonthaburi

OFFICIAL RECEIPT

Date: April 20, 2026
Receipt No: RCT26-00005
Payment Method: Cash

INVOICE NO	AMOUNT DUE	AMT RECEIVED HERE	STILL REMAINING
INV-032	฿2,096.93	฿2,096.93	฿0.00
INV-038	฿2,573.89	฿2,573.89	฿0.00
INV-050	฿1,221.67	฿1,221.67	฿0.00

Total Received: ฿5,892.49

ภาพหน้าจอแสดงหน้าจอใบเสร็จรับเงินที่ถูกพิมพ์แล้ว (Printed receipt) โดยเลือกใบเสร็จที่มีรายการย่อย 2–3 รายการ จำนวน 1 ใบเสร็จ

5. Show Report 1 (List of receipts) with filter inputs and report output is working.

List of Receipts

All received customer payments

Filters 2

Customer

C102 - Siam Trading Co., Ltd.

LoV

×

Date Range

วว/มม./ปปปป

to

วว/มม./ปปปป

Reset

Apply Filters

Select your filters

Choose filter options above and click "Apply Filters" to generate the report.

List of Receipts

All received customer payments

Filters 2

Customer

C102 - Siam Trading Co., Ltd.

LoV

×

Date Range

วว/มม./ปปปป

to

วว/มม./ปปปป

Reset

Apply Filters

1 total records

Refresh

RECEIPT NO. ▲	DATE ▲	CUSTOMER ▲	PAYMENT METHOD ▲	PAYMENT NOTES	TOTAL AMOUNT RECEIVED ▲
RCT26-00003	April 21, 2026	Siam Trading Co., Ltd. (C102)	Check		\$9,548.43

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ภาพหน้าจอแสดงรายงานที่ 1 (รายการใบเสร็จรับเงิน หรือ List of Receipts) โดยให้ตัวกรองข้อมูล (filter inputs) และผลลัพธ์รายงาน (report output) ว่าทำงานได้อย่างถูกต้อง

6. Show Report 2 (List of invoices and receipt information) with filter input and report output is working.

List of Invoices and Receipt Information

Invoice balances and received amount details

Customer

C102 - Siam Trading Co., Ltd.

LoV

×

Date Range

วว/มม./ปปปป

to

วว/มม./ปปปป

Reset

Apply Filters

Select your filters

Choose filter options above and click "Apply Filters" to generate the report.

List of Invoices and Receipt Information

Invoice balances and received amount details

Customer

C102 - Siam Trading Co., Ltd.

LoV

×

Date Range

วว/มม./ปปปป

to

วว/มม./ปปปป

Reset

Apply Filters

5 total records

Refresh

INVOICE NO. ▲	INV DATE ▲	CUSTOMER ▲	AMOUNT DUE ▲	AMT RECEIVED ▲	AMT REMAINING ▲	RECEIPT NO. ▲	RECEIPT DATE	RECEIVED HERE
INV-002	January 03, 2026	Siam Trading Co., Ltd. (C102)	\$2,169.43	\$2,169.43	\$0.00	RCT26-00003	April 21, 2026	\$2,169.43
INV-013	December 22, 2025	Siam Trading Co., Ltd. (C102)	\$1,407.32	\$1,407.32	\$0.00	RCT26-00003	April 21, 2026	\$1,407.32
INV-027	November 25, 2025	Siam Trading Co., Ltd. (C102)	\$390.02	\$390.02	\$0.00	RCT26-00003	April 21, 2026	\$390.02
INV-035	December 20, 2025	Siam Trading Co., Ltd. (C102)	\$1,000.32	\$1,000.32	\$0.00	RCT26-00003	April 21, 2026	\$1,000.32
INV-040	December 26, 2025	Siam Trading Co., Ltd. (C102)	\$4,581.34	\$4,581.34	\$0.00	RCT26-00003	April 21, 2026	\$4,581.34

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ภาพหน้าจอแสดงรายงานที่ 2 (รายการใบแจ้งหนี้และข้อมูลใบเสร็จรับเงิน หรือ List of Invoices and Receipt Information) โดยให้ตัวกรองข้อมูล (filter input) และผลลัพธ์รายงาน (report output) ว่าทำงานได้อย่างถูกต้อง

2. Code Submission

1) Showing invoice_lab4_delta.sql that must include all schema changes from Individual Lab 3 to Invoice Lab 4

นี่คือคำสั่ง SQL สำหรับอัปเดต Database โดยไม่มีการ Drop ของเดิมทิ้ง ซึ่งจะมีคำสั่งที่อยู่ในไฟล์ `invoice_lab4_delta.sql` สำหรับปรับปรุง Database มีรายละเอียดดังนี้

```
-- 1. Create Receipt Header Table
CREATE TABLE IF NOT EXISTS receipt (
    id SERIAL PRIMARY KEY,
    receipt_no VARCHAR(20) UNIQUE NOT NULL,
    receipt_date DATE NOT NULL,
    customer_id INTEGER NOT NULL REFERENCES customer(id),
    payment_method VARCHAR(50) CHECK (payment_method IN ('Cash', 'Bank
Transfer', 'Check')),
    payment_notes TEXT,
    total_received NUMERIC(12,2) NOT NULL DEFAULT 0.00,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 2. Create Receipt Line Item Table
CREATE TABLE IF NOT EXISTS receipt_line_item (
    id SERIAL PRIMARY KEY,
    receipt_id INTEGER NOT NULL REFERENCES receipt(id) ON DELETE CASCADE,
    invoice_id INTEGER NOT NULL REFERENCES invoice(id),
    amount_received NUMERIC(12,2) NOT NULL DEFAULT 0.00,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 3. Create invoice_received_view View
-- This view calculates the total amount received and the remaining amount
for each invoice.
DROP VIEW IF EXISTS invoice_received_view;
CREATE VIEW invoice_received_view AS
SELECT
    i.customer_id,
    i.id AS invoice_id,
    i.invoice_no,
    i.amount_due,
    COALESCE(SUM(rli.amount_received), 0.00) AS amount_received,
    i.amount_due - COALESCE(SUM(rli.amount_received), 0.00) AS amount_remain
```

```
FROM
    invoice i
LEFT JOIN
    receipt_line_item rli ON i.id = rli.invoice_id
GROUP BY
    i.customer_id, i.id, i.invoice_no, i.amount_due;
```


2) Frontend Code: Receipt Form & Related APIs

- **client/src/api/receipts.api.js (New File):** สร้างไฟล์ API ใหม่สำหรับการเรียก Backend ทุก endpoint ที่เกี่ยวข้องกับ Receipt ได้แก่ list, get, create, update, delete และ getUnpaidInvoices

```
import { API_BASE } from "../http.js";

// API for Receipts
export async function listReceipts(params = {}) {
  const q = new URLSearchParams(params).toString();
  const res = await fetch(`${API_BASE}${API_BASE.endsWith('/') ? '' : '/'}api/receipts?${q}`);
  if (!res.ok) { const text = await res.text(); throw new Error(text || "Failed to fetch receipts"); }
  return res.json();
}

export async function getReceipt(id) {
  if (id === "new") return null;
  const res = await fetch(`${API_BASE}${API_BASE.endsWith('/') ? '' : '/'}api/receipts/${id}`);
  if (!res.ok) { const text = await res.text(); throw new Error(text || "Failed to fetch receipt"); }
  return res.json();
}

export async function createReceipt(data) {
  const res = await fetch(`${API_BASE}${API_BASE.endsWith('/') ? '' : '/'}api/receipts`, {
    method: "POST",
    headers: { "Content-Type": "application/json" },
    body: JSON.stringify(data),
  });
  if (!res.ok) {
    let err;
    try { const text = await res.text(); err = JSON.parse(text).error; }
    catch { err = "Create error"; }
    throw new Error(err);
  }
  return res.json();
}
```

```

}

export async function updateReceipt(id, data) {
  const res = await fetch(`${API_BASE}${API_BASE.endsWith('/') ? '' :
  '/'}api/receipts/${id}`, {
    method: "PUT",
    headers: { "Content-Type": "application/json" },
    body: JSON.stringify(data),
  });
  if (!res.ok) {
    let err;
    try { const text = await res.text(); err = JSON.parse(text).error; }
  catch { err = "Update error"; }
    throw new Error(err);
  }
  return res.json();
}

export async function deleteReceipt(id) {
  const res = await fetch(`${API_BASE}${API_BASE.endsWith('/') ? '' :
  '/'}api/receipts/${id}`, { method: "DELETE" });
  if (!res.ok) { const text = await res.text(); throw new Error(text ||
  "Failed to delete receipt"); }
  return res.json();
}

export async function getReceiptUnpaidInvoices(customerCode, excludeReceiptId
= "") {
  const params = { customerCode };
  if (excludeReceiptId) params.excludeReceiptId = excludeReceiptId;
  const q = new URLSearchParams(params).toString();
  const res = await fetch(`${API_BASE}${API_BASE.endsWith('/') ? '' :
  '/'}api/receipt-invoices?${q}`);
  if (!res.ok) {
    let err;
    try { const text = await res.text(); err = JSON.parse(text).error; }
  catch { err = "Fetch error"; }
    throw new Error(err || "Failed to fetch unpaid invoices");
  }
  return res.json();
}

```

- client/src/pages/receipts/ReceiptList.jsx (New File): สร้างหน้า List Receipts โดยใช้ DataList Component เดียวกับ InvoiceList และมี Confirm Modal สำหรับลบ

```
import React from "react";
import { toast } from "react-toastify";
import { listReceipts, deleteReceipt } from "../../api/receipts.api.js";
import { formatBaht, formatDate } from "../../utils.js";
import DataList from "../../components/DataList.jsx";
import { ConfirmModal, AlertModal } from "../../components/Modal.jsx";

export default function ReceiptList() {
  const fetchData = React.useCallback((params) => listReceipts(params), []);
  const [confirmModal, setConfirmModal] = React.useState({ isOpen: false,
id: null });
  const [alertModal, setAlertModal] = React.useState({ isOpen: false,
message: "" });
  const [refreshTrigger, setRefreshTrigger] = React.useState(0);

  const closeConfirm = () => setConfirmModal({ isOpen: false, id: null });

  const handleDelete = (id) => {
    setConfirmModal({ isOpen: true, id });
  };

  const confirmDelete = async () => {
    try {
      await deleteReceipt(confirmModal.id);
      closeConfirm();
      setRefreshTrigger((t) => t + 1);
      toast.success("Receipt deleted.");
    } catch (e) {
      const msg = String(e.message || e);
      toast.error(msg);
      setAlertModal({ isOpen: true, message: "Error: " + msg });
    }
  };

  const columns = [
    { key: "receipt_no", label: "Receipt No", sortable: true },
```

```

        { key: "receipt_date", label: "Date", render: (v) => formatDate(v),
sortable: true },
        { key: "customer_name", label: "Customer Name", sortable: true },
        { key: "payment_method", label: "Payment Method", sortable: true },
        { key: "total_received", label: "Total Received", align: "right",
render: (v) => formatBaht(v), sortable: true },
    ];

    return (
        <div>
            <DataList
                title="Receipts"
                basePath="/receipts"
                itemName="receipt"
                itemKey="receipt_no"
                columns={columns}
                fetchData={fetchData}
                onDelete={handleDelete}
                refreshTrigger={refreshTrigger}
                placeholder="Search by receipt no or customer name..."
            />
            <ConfirmModal
                isOpen={confirmModal.isOpen}
                onClose={closeConfirm}
                onConfirm={confirmDelete}
                closeOnConfirm={false}
                title="Delete Receipt"
                message="Are you sure you want to delete this receipt?"
                confirmText="Delete"
            />
            <AlertModal
                isOpen={alertModal.isOpen}
                onClose={() => setAlertModal({ isOpen: false, message: "" })}
                title="Error"
                message={alertModal.message}
            />
        </div>
    );
}

```

- `client/src/pages/receipts/ReceiptPage.jsx` (New File): สร้างหน้าฟอร์ม Receipt ที่รองรับ 3 Mode: Create / View / Edit พร้อมระบบ Line-item LoV สำหรับเลือก Unpaid Invoice และคำนวณ Amount Remaining แบบ Real-time
- ส่วนที่ 1 : Header State & Customer Picker:

```
export default function ReceiptPage({ mode: propMode }) {
  const { id } = useParams();
  const mode = propMode || (id ? "view" : "create");
  // Header state
  const [receiptDate, setReceiptDate] = React.useState(
    new Date().toISOString().split("T")[0]
  );
  const [paymentMethod, setPaymentMethod] = React.useState("Cash");
  const [customerCode, setCustomerCode] = React.useState("");
  const [customerName, setCustomerName] = React.useState("");
  const [lines, setLines] = React.useState([]);
  // ...
}
```

- ส่วนที่ 2 : Logic คำนวณ Amount Remaining แบบ Real-time:

```
const updateLineAmount = (id, newAmountText) => {
  const newVal = Number(newAmountText) || 0;
  setLines(lines.map(li => {
    if (li.id === id) {
      return {
        ...li,
        amount_received_here: newAmountText,
        // คำนวณ Still Remaining หักที่ที่พิมพ์
        amount_still_remaining: round2(li.amount_remaining - newVal)
      };
    }
    return li;
  }));
};
```

- ส่วนที่ 3 : ดึง Unpaid Invoices จาก Backend (Invoice LoV):

```
const loadUnpaidInvoices = async () => {
  if (!customerCode) {
    toast.warning("Please select a customer first.");
    return;
  }
  try {
    // เรียก API เพื่อดึงเฉพาะ Invoice ที่ยังค้างชำระของลูกค้านั้น
    const data = await getReceiptUnpaidInvoices(customerCode, receiptId || null);
    setUnpaidInvoices(data);
    setInvoiceLovOpen(true);
  } catch (e) {
    toast.error(e.message);
  }
};
```

- **ส่วนที่ 4 :** Payment Method Dropdown ต้องตรงกับ DB CHECK Constraint:

```
<select
  required
  className="form-control"
  value={paymentMethod}
  onChange={(e) => setPaymentMethod(e.target.value)}
>
  <option value="Cash">Cash</option>
  <option value="Bank Transfer">Bank Transfer</option>
  <option value="Check">Check</option>
</select>
```

- **ส่วนที่ 5 :** Print CSS ซ่อนคอลัมน์ที่ไม่จำเป็นตอน Print:

```
<style>{`
  @media print {
    .receipt-print-hide { display: none !important; }
    .card { box-shadow: none !important; border: none !important; }
    .modern-table { font-size: 12px; }
    .page-header, .no-print { display: none !important; }
  }
`}</style>
```

3) Backend Code: Receipt Service & Controller & Routes

- `server/src/services/receipts.service.js` (New File): สร้าง Service ใหม่ทั้งหมดสำหรับ Receipt โดยมี Logic สำคัญดังนี้
- ส่วนที่ 1 : Generate receipt_no แบบ Auto Serial (RCTYY-XXXXX):

```
// Generate RCTYY-XXXXX (e.g. RCT26-00001)
const maxRes = await client.query("SELECT MAX(id) as m FROM receipt");
const nextId = (maxRes.rows[0].m || 0) + 1;
const year = new Date(receipt_date ||
Date.now()).getFullYear().toString().slice(-2);
const receipt_no = `RCT${year}-${nextId.toString().padStart(5, "0")}`;
```

- ส่วนที่ 2 : createReceipt: INSERT Header + Line Items ในธุรกรรมเดียว:

```
export async function createReceipt({ receipt_date, customer_code,
payment_method, payment_notes, line_items }) {
  const client = await pool.connect();
  try {
    await client.query("begin");

    // Lookup customer_id จาก customer_code
    const cust = await client.query("SELECT id FROM customer WHERE code = $1",
[customer_code]);
    if (cust.rowCount === 0) throw new Error(`Customer not found:
${customer_code}`);
    const customer_id = cust.rows[0].id;

    // [Generate receipt_no ตามข้างบน]

    // INSERT receipt header
    const rct = await client.query(
      `INSERT INTO receipt (id, receipt_no, receipt_date, customer_id,
payment_method, payment_notes, total_received, created_at)
      VALUES ((SELECT COALESCE(MAX(id),0)+1 FROM receipt), $1, $2, $3, $4,
$5, $6, NOW())
      RETURNING id, receipt_no`,
      [receipt_no, receipt_date, customer_id, payment_method, payment_notes,
round2(total_received)]
    );
```

```

const receipt_id = rct.rows[0].id;

// INSERT line items
for (const li of validLines) {
  await client.query(
    `INSERT INTO receipt_line_item (id, receipt_id, invoice_id,
amount_received, created_at)
    VALUES ((SELECT COALESCE(MAX(id),0)+1 FROM receipt_line_item), $1,
$2, $3, NOW())`,
    [receipt_id, li.invoice_id, li.amount_received]
  );
}

await client.query("commit");
return { receipt_no: rct.rows[0].receipt_no };
} catch (err) {
  await client.query("rollback");
  throw err;
} finally {
  client.release();
}
}

```

- ส่วนที่ 3 : ดึง Unpaid Invoices จาก Backend (Invoice LoV):

```

export async function getUnpaidInvoices(customerCode, excludeReceiptId = null)
{
  const cust = await pool.query("SELECT id FROM customer WHERE code = $1",
[customerCode]);
  if (cust.rowCount === 0) throw new Error(`Customer not found:
${customerCode}`);
  const customerId = cust.rows[0].id;

  const query = `
    SELECT v.invoice_id, v.invoice_no, v.amount_due as full_amount_due,
      COALESCE((SELECT SUM(amount_received) FROM receipt_line_item
        WHERE invoice_id = v.invoice_id
        AND ($2::int IS NULL OR receipt_id != $2::int)), 0.00) as
amount_already_received
    FROM invoice_received_view v
  `

```



```

WHERE v.customer_id = $1
AND (
  -- คงเหลือหลังหักยอดชำระก่อนหน้า > 0
  (v.amount_due - COALESCE((SELECT SUM(amount_received) FROM
receipt_line_item
                                WHERE invoice_id = v.invoice_id
                                AND ($2::int IS NULL OR receipt_id !=
$2::int)), 0.00)) > 0
  -- หรือ Invoice นี้อยู่ใน Receiptที่กำลัง edit อยู่ (สำหรับ Edit Mode)
  OR v.invoice_id IN (SELECT invoice_id FROM receipt_line_item WHERE
receipt_id = $2::int)
)
ORDER BY v.invoice_no ASC
`;
const res = await pool.query(query, [customerId, excludeReceiptId || null]);

return res.rows.map(r => ({
  invoice_id: Number(r.invoice_id),
  invoice_no: r.invoice_no,
  full_amount_due: Number(r.full_amount_due),
  amount_already_received: Number(r.amount_already_received),
  amount_remaining: round2(Number(r.full_amount_due) -
Number(r.amount_already_received))
})));
}

```

- `server/src/services/receipts.controller.js (New File)`: สร้าง Controller ใหม่สำหรับ Receipt รับ HTTP request แล้วส่งต่อไปยัง Service

```
import * as receiptService from "../services/receipts.service.js";

export async function getReceipts(req, res) {
  try {
    const result = await receiptService.listReceipts(req.query);
    res.json(result);
  } catch (err) {
    res.status(500).json({ error: "Internal Server Error" });
  }
}

export async function getReceiptById(req, res) {
  try {
    const result = await receiptService.getReceipt(req.params.id);
    if (!result) return res.status(404).json({ error: "Receipt not found" });
    res.json(result);
  } catch (err) {
    res.status(500).json({ error: "Internal Server Error" });
  }
}

export async function createReceipt(req, res) {
  try {
    const result = await receiptService.createReceipt(req.body);
    res.status(201).json(result);
  } catch (err) {
    res.status(400).json({ error: err.message });
  }
}

export async function updateReceipt(req, res) {
  try {
    const result = await receiptService.updateReceipt(req.params.id, req.body);
    res.json(result);
  } catch (err) {
```

```

        res.status(400).json({ error: err.message });
    }
}

export async function deleteReceipt(req, res) {
    try {
        await receiptService.deleteReceipt(req.params.id);
        res.json({ ok: true });
    } catch (err) {
        res.status(500).json({ error: err.message });
    }
}

// Invoice LoV: ดึง Unpaid Invoices ของลูกค้าที่เลือก
export async function getReceiptUnpaidInvoices(req, res) {
    try {
        const customerCode = req.query.customerCode;
        let currentReceiptId = req.query.excludeReceiptId;
        // กัน string "null" ที่ส่งมาจาก URLSearchParams
        if (currentReceiptId === "null" || currentReceiptId === "")
            currentReceiptId = null;
        if (!customerCode) return res.status(400).json({ error: "customerCode is required" });
        const result = await receiptService.getUnpaidInvoices(customerCode, currentReceiptId);
        res.json(result);
    } catch (err) {
        res.status(400).json({ error: err.message });
    }
}

```

- `server/src/services/receipts.routes.js` (New File): ลงทะเบียน Route ทั้งหมดสำหรับ Receipt รวมถึง endpoint สำหรับ Invoice LoV

```
import express from "express";
import * as receiptsController from "../controllers/receipts.controller.js";

const router = express.Router();

router.get("/api/receipt-invoices",
receiptsController.getReceiptUnpaidInvoices);

router.get("/api/receipts", receiptsController.getReceipts);
router.post("/api/receipts", receiptsController.createReceipt);

router.get("/api/receipts/:id", receiptsController.getReceiptById);
router.put("/api/receipts/:id", receiptsController.updateReceipt);
router.delete("/api/receipts/:id", receiptsController.deleteReceipt);

export default router;
```

4) Report Code: Backend & Frontend

- `server/src/services/reports.service.js` (Modified File): เพิ่ม Function `getReceiptListReport` และ `getInvoiceReceiptsReport` ต่อท้ายไฟล์เดิม

- ส่วนที่ 1 : `getReceiptListReport`: ดึงรายการ Receipt พร้อม Filter:

```
export async function getReceiptListReport({
  date_from, date_to, customer_code,
  page = 1, limit = 10, sortBy = "receipt_date", sortDir = "desc",
} = {}) {
  // [resolve customer_id จาก customer_code ถ้ามี]

  let whereClause = "WHERE 1=1";
  const params = [];
  let paramIndex = 1;

  if (resolvedCustomerId) {
    whereClause += ` AND c.id = ${paramIndex++}`;
    params.push(resolvedCustomerId);
  }
  if (date_from) {
    whereClause += ` AND r.receipt_date >= ${paramIndex++}`;
    params.push(date_from);
  }
  if (date_to) {
    whereClause += ` AND r.receipt_date <= ${paramIndex++}`;
    params.push(date_to);
  }

  const { rows } = await pool.query(
    `SELECT r.receipt_no, r.receipt_date, c.code as customer_code, c.name as
customer_name,
        r.payment_method, r.payment_notes, r.total_received
FROM receipt r
JOIN customer c ON c.id = r.customer_id
${whereClause}
ORDER BY ${sortBy} ${sortDir}
LIMIT ${paramIndex++} OFFSET ${paramIndex++}`,
    [...params, Number(limit), offset]
  );
}
```

```
    return { data: rows, total, page: Number(page), limit: Number(limit),
totalPages };
}
```

- **ส่วนที่ 2 :** getInvoiceReceiptsReport: ดึงข้อมูล Invoice พร้อมยอดรับชำระจาก invoice_received_view:

```
export async function getInvoiceReceiptsReport({
  date_from, date_to, customer_code,
  page = 1, limit = 10, sortBy = "invoice_no", sortDir = "desc",
} = {}) {
  // [build WHERE clause เหมือนกัน]

  const { rows } = await pool.query(
    `SELECT v.invoice_no, i.invoice_date, c.code as customer_code, c.name as
customer_name,
          v.amount_due, v.amount_received as invoice_amount_received,
v.amount_remain,
          r.receipt_no, r.receipt_date, rli.amount_received as
receipt_amount
FROM invoice_received_view v
JOIN customer c ON c.id = v.customer_id
JOIN invoice i ON i.id = v.invoice_id
LEFT JOIN receipt_line_item rli ON rli.invoice_id = i.id
LEFT JOIN receipt r ON r.id = rli.receipt_id
${whereClause}
ORDER BY ${sortColumn} ${sortDirection}, r.receipt_date ASC
LIMIT ${paramIndex++} OFFSET ${paramIndex++}` ,
    [...params, Number(limit), offset]
  );
  return { data: rows, total, page: Number(page), limit: Number(limit),
totalPages };
}
```

- `server/src/services/reports.controller.js` (Modified File): เพิ่ม Controller 2 ตัวสำหรับ Report ใหม่

```
export async function getReceiptListReport(req, res) {
  try {
    const result = await reportsService.getReceiptListReport(req.query);
    sendList(res, result);
  } catch (err) {
    sendError(res, err?.message ?? String(err), 500);
  }
}

export async function getInvoiceReceiptsReport(req, res) {
  try {
    const result = await reportsService.getInvoiceReceiptsReport(req.query);
    sendList(res, result);
  } catch (err) {
    sendError(res, err?.message ?? String(err), 500);
  }
}
```

- `server/src/services/reports.routes.js` (Modified File): เพิ่ม Route สำหรับ Report ใหม่ 2 เส้นทาง

```
r.get("/receipt-list", c.getReceiptListReport);
r.get("/invoice-receipts", c.getInvoiceReceiptsReport);
```

- `server/src/services/reports.api.js` (Modified File): เพิ่ม Endpoint ใหม่ 2 รายการใน `REPORT_ENDPOINTS` Map

```
const REPORT_ENDPOINTS = {
  "product-sales": "/api/reports/product-sales",
  "monthly-sales": "/api/reports/product-monthly-sales",
  "customer-sales": "/api/reports/customer-sales",
  "customer-outstanding": "/api/reports/customer-outstanding",
  // เพิ่มใหม่สำหรับ Individual 4
  "receipt-list": "/api/reports/receipt-list",
  "invoice-receipts": "/api/reports/invoice-receipts",
};
```

- client/src/pages/reports/Reports.jsx (Modified File): เพิ่ม
REPORT_CONFIG สำหรับ Report ใหม่ 2 ตัว พร้อมกำหนดคอลัมน์ตาราง

```
"receipt-list": {
  title: "List of Receipts",
  subtitle: "All received customer payments",
  emptyMessage: "No receipt records found.",
  getColumns: () => [
    { key: "receipt_no", label: "Receipt No.", sortable: true,
      render: (v) => <span className="font-bold">{v}</span> },
    { key: "receipt_date", label: "Date", sortable: true, render: (v) =>
      formatDate(v) },
    { key: "customer_code", label: "Customer", sortable: true,
      render: (_, row) => `${row.customer_name} (${row.customer_code})` },
    { key: "payment_method", label: "Payment Method", sortable: true },
    { key: "payment_notes", label: "Payment Notes" },
    { key: "total_received", label: "Total Amount Received", align: "right",
      sortable: true,
      render: (v) => formatBaht(v) }
  ]
},
"invoice-receipts": {
  title: "List of Invoices and Receipt Information",
  subtitle: "Invoice balances and received amount details",
  emptyMessage: "No invoice/receipt records found.",
  getColumns: () => [
    { key: "invoice_no", label: "Invoice No.", sortable: true,
      render: (v) => <span className="font-bold">{v}</span> },
    { key: "invoice_date", label: "Inv Date", sortable: true, render: (v) =>
      formatDate(v) },
    { key: "customer_code", label: "Customer", sortable: true,
      render: (_, row) => `${row.customer_name} (${row.customer_code})` },
    { key: "amount_due", label: "Amount Due", align: "right", sortable:
      true,
      render: (v) => formatBaht(v) },
    { key: "invoice_amount_received", label: "Amt Received", align: "right",
      sortable: true,
      render: (v) => formatBaht(v) },
    { key: "amount_remain", label: "Amt Remaining", align: "right",
      sortable: true,
```



```

        style: { color: "var(--danger)" }, render: (v) => formatBaht(v) },
        { key: "receipt_no", label: "Receipt No.", sortable: true },
        { key: "receipt_date", label: "Receipt Date", render: (v) => v ?
formatDate(v) : "-" },
        { key: "receipt_amount", label: "Received Here", align: "right",
render: (v) => v ? formatBaht(v) : "-" }
    ]
}

```

- [client/src/pages/reports/filters/ReportFilters.jsx](#) (Modified File): เพิ่ม Filter Block สำหรับ Report ใหม่ทั้ง 2 ตัว (Customer Filter + Date Range Filter)

```

{/* Receipt List Filters */}
{type === "receipt-list" && (
    <>
        <CustomerFilter value={filters.customerCode || ""}
            displayLabel={filters.customerLabel}
            onChange={handleChange} />
        <DateRangeFilter dateFrom={filters.dateFrom || ""}
            dateTo={filters.dateTo || ""}
            onChange={handleChange} />
    </>
)}

{/* Invoice Receipts Filters */}
{type === "invoice-receipts" && (
    <>
        <CustomerFilter value={filters.customerCode || ""}
            displayLabel={filters.customerLabel}
            onChange={handleChange} />
        <DateRangeFilter dateFrom={filters.dateFrom || ""}
            dateTo={filters.dateTo || ""}
            onChange={handleChange} />
    </>
)}

```

- `client/src/main.jsx` (Modified File): เพิ่ม Sidebar Navigation Link และ Route สำหรับ Receipts และ Report ใหม่ทั้ง 2 ตัว

- เพิ่ม NavLink ใน Sidebar (Reports Sub-menu):

```
<NavLink to="/reports/receipt-list" className={getSubLinkClass}>
  List of Receipts
</NavLink>
<NavLink to="/reports/invoice-receipts" className={getSubLinkClass}>
  Invoices & Receipts
</NavLink>
```

- เพิ่ม Route สำหรับ Receipt Pages:

```
<Route path="/receipts" element={<Layout><ReceiptList /></Layout>} />
<Route path="/receipts/new" element={<Layout><ReceiptPage mode="create"
/></Layout>} />
<Route path="/receipts/:id" element={<Layout><ReceiptPage mode="view"
/></Layout>} />
<Route path="/receipts/:id/edit" element={<Layout><ReceiptPage mode="edit"
/></Layout>} />
```

- เพิ่ม Route สำหรับ Report ใหม่:

```
<Route path="/reports/receipt-list"
  element={<Layout><Reports type="receipt-list" /></Layout>} />
<Route path="/reports/invoice-receipts"
  element={<Layout><Reports type="invoice-receipts" /></Layout>} />
```